

Table: EmployeeProjects

EmpID	EmpName	DeptID	DeptName	ProjectID	ProjectName	ProjectBudget	ManagerID	ManagerName
E01	Areeba	D10	HR	P101	Recruitment	50000	M01	Zara Khan
E01	Areeba	D10	HR	P102	Training	30000	M01	Zara Khan
E02	Hamna	D20	IT	P103	Automation	80000	M02	Bilal Ahmed
E02	Hamna	D20	IT	P104	Analytics	90000	M02	Bilal Ahmed
E03	Noor	D10	HR	P101	Recruitment	50000	M01	Zara Khan

🔑 **Primary Key: EmpID + ProjectID**

Functional Dependencies:

- EmpID → EmpName, DeptID
- DeptID → DeptName
- ProjectID → ProjectName, ProjectBudget, ManagerID
- ManagerID → ManagerName

2NF Decomposition

Table Name	Attributes
WorksOn	EmpID, ProjectID
Employee	EmpID, EmpName, DeptID
Project	ProjectID, ProjectName, ProjectBudget, ManagerID

3NF Decomposition

We remove transitive dependencies now:

Table Name	Attributes
WorksOn	EmpID, ProjectID
Employee	EmpID, EmpName, DeptID
Department	DeptID, DeptName
Project	ProjectID, ProjectName, ProjectBudget, ManagerID
Manager	ManagerID, ManagerName

Total Tables in 3NF: 5

Table: UniversityCourseRegistration

RegNo	StudentName	Email	CourseID	CourseName	CreditHours	InstructorID	InstructorName	InstructorEmail	DeptID	DeptName
R101	Hamza	hamza@tuf.edu.pk	C101	DBMS	3	I10	Dr. Farooq	farooq@tuf.edu.pk	D1	CS
R102	Areeba	areeba@tuf.edu.pk	C102	AI	4	I11	Dr. Sana	sana@tuf.edu.pk	D1	CS
R103	Hamna	hamna@tuf.edu.pk	C103	Psychology	2	I12	Dr. Fatima	fatima@tuf.edu.pk	D2	Humanities
R101	Hamza	hamza@tuf.edu.pk	C102	AI	4	I11	Dr. Sana	sana@tuf.edu.pk	D1	CS

🔑 **Primary Key: RegNo + CourseID**

Functional Dependencies:

- RegNo → StudentName, Email
- CourseID → CourseName, CreditHours, InstructorID
- InstructorID → InstructorName, InstructorEmail, DeptID
- DeptID → DeptName

2NF Decomposition

Table Name	Attributes
Registration	RegNo, CourseID
Student	RegNo, StudentName, Email
Course	CourseID, CourseName, CreditHours, InstructorID

Total Tables in 2NF: 3

3NF Decomposition

We now remove transitive dependencies:

Table Name	Attributes
Registration	RegNo, CourseID
Student	RegNo, StudentName, Email
Course	CourseID, CourseName, CreditHours, InstructorID
Instructor	InstructorID, InstructorName, InstructorEmail, DeptID
Department	DeptID, DeptName

Total Tables in 3NF: 5